

Class Test II: Advanced Operating Systems
MSc Semester II, 2024

Full Marks: 20

Answer all the Questions

Time: 45 minutes

2x4=8

Q1. Answer the following (within 200 words for each question):

- a. State at least two different motivations behind process migration.
- b. Why token based algorithms are said to be inherently safe?
- c. Define pre-emptive and non pre-emptive process migration.
- d. How do you interpret liveness and safety when you are designing an algorithm to detect possible deadlock in a system?

Q2. Comment on the correctness of the statement and justify your opinion: (any three):

4x3=12

- a. Symmetric algorithms involve lower communication overhead in comparison with the diffusion computation approach.
- b. Raymond's algorithm may grant access to processes out of turn.
- c. In a distributed system, resource migration is more challenging than migration of codes for a process.
- d. Receiver-initiated process migration requires support for pre-emptive migration.